

04 · Which funnel stage to invest in? — mediation (pathmc)

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The business decision. We have a fixed budget to split across the growth funnel — **onboarding** -> **engagement** -> **activation** -> **conversion** — and want to put it where it moves paid conversion the most. The complication is that the stages form a **chain**: money spent improving onboarding doesn't only help onboarding, it *cascades* downstream into engagement, activation, and ultimately conversion. So we can't just correlate each stage with revenue; we have to trace the causal *paths* and decide where a push travels furthest.

1.0.1 The concept: mediation, direct and indirect effects

When a treatment X affects an outcome Y *through* an intermediate variable M (a **mediator**), the total effect splits in two:

- the **direct effect** — X 's influence on Y that does *not* go through M (Pearl's *natural direct effect*, NDE);
- the **indirect effect** — the part that flows $X \rightarrow M \rightarrow Y$ (the *natural indirect effect*, NIE).

Total effect = direct + indirect, and the **proportion mediated** = indirect / total tells you how much of a stage's value is really realised *downstream*. In a linear model the indirect effect is just the **product of the coefficients along the path** (the classic " $a \cdot b$ " of mediation analysis) — **pathmc** generalises this by pushing posterior draws through the intervened graph, so we get every path's effect *with uncertainty*.

1.0.2 The catch this notebook is honest about

Mediation leans on a **stronger** assumption than a simple treatment-effect analysis: no unmeasured confounding of the *mediator-outcome* link (on top of the usual treatment-outcome one). If some unmeasured trait drives both engagement *and* conversion, the direct/indirect *split* is biased even when the *total* effect is fine. So Step 6 doesn't just report the split — it **stress-tests how much such confounding would have to exist to flip the "invest here" decision**.

On real data. Swap in your **product-analytics event data** — per-user onboarding score, engagement, activation flag, conversion — which every growth team already logs. The method needs the funnel's causal *order* to be known (it usually is) and the no-hidden-mediator-confounder assumption to be defensible (Step 6 shows how to argue it).

pathmc builds the chain as a structural causal model and computes **path-specific effects** by propagating posterior draws through the intervened graph — the modern, uncertainty-aware mediation

analysis. 7-step contract.

1.1 2 · Simulate a ground truth

A chain `onboarding_score -> engagement -> activated -> converted`, with `channel_quality` feeding engagement and activation. The planted structure gives a known **total** effect of onboarding on conversion, split into paths — most flows through **engagement** -> **activated**, a little through **engagement** -> **converted** directly.

TRUE effects of onboarding on conversion:

indirect_via_engagement_activated	3.17
indirect_via_engagement_only	0.77
total	3.94

	onboarding_score	channel_quality	engagement	activated	converted
0	0.864798	0.726815	2.576471	3.869429	4.870321
1	0.855303	0.790067	3.206651	5.673260	6.371102
2	0.811023	0.348493	2.493785	4.691864	5.394113
3	0.261446	0.465793	0.572682	1.261547	1.264003
4	0.077199	0.887974	1.520775	3.259916	3.146544

1.2 3 · Identify — direct vs indirect effects (NDE / NIE)

For a chain $X \rightarrow M \rightarrow Y$ the **total effect** decomposes into a **natural direct effect** and **natural indirect effect** (Pearl):

$$\text{TE} = \text{NDE} + \text{NIE}, \quad \text{NDE} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))], \quad \text{NIE} = \mathbb{E}[Y(1, M(1)) - Y(1, M(0))].$$

Under a linear SCM these are the **product-of-coefficients**: indirect = $a \cdot b$, direct = c . The **proportion mediated** = NIE/TE tells you how much of onboarding’s value is “really” downstream engagement/activation. `pathmc` computes path-specific effects for the full chain by simulating through the intervened graph.

Assumptions (stronger than a single-link analysis): no unmeasured confounding of $X-Y$, $M-Y$, and $X-M$, plus the **cross-world** condition that no $M-Y$ confounder is itself affected by X . Natural effects are *not identified* if that last one fails — and the $M-Y$ confounding assumption is the one most likely to break, so we stress-test it in step 6.

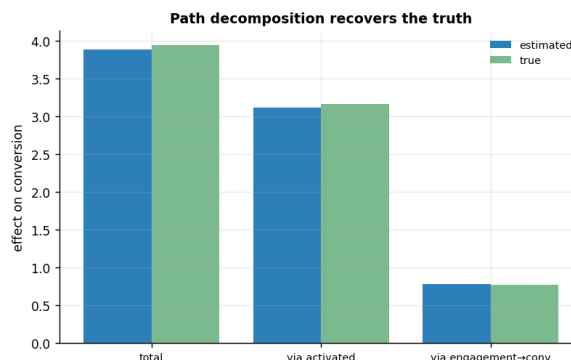
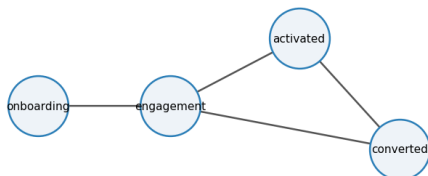
1.3 4 · Estimate — fit the chain and query every path

	mean	sd	hdi_3%	hdi_97%
name				
e_on	2.150	0.042	2.073	2.226
e_ch	1.131	0.043	1.043	1.207
a_eng	1.616	0.018	1.584	1.652
a_ch	0.592	0.054	0.490	0.690
c_act	0.896	0.022	0.857	0.940
c_eng	0.361	0.040	0.286	0.430

1.4 5 · Validate — recover the total, the paths, and the proportion mediated

TOTAL onboarding → converted 3.89 (true 3.94) [90% 3.75, 4.03]
 via engagement→activated→conv 3.11 (true 3.17) [90% 2.95, 3.28]
 via engagement→converted 0.78 (true 0.77) [90% 0.63, 0.92]
 Proportion mediated (indirect/total): 100% [99%, 101%] – most of onboarding's value is realised downstream.

The funnel chain (paths onboarding → converted)

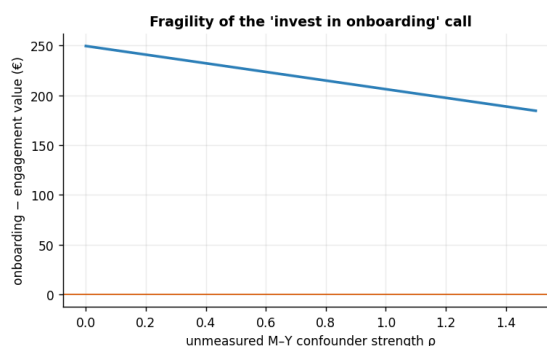
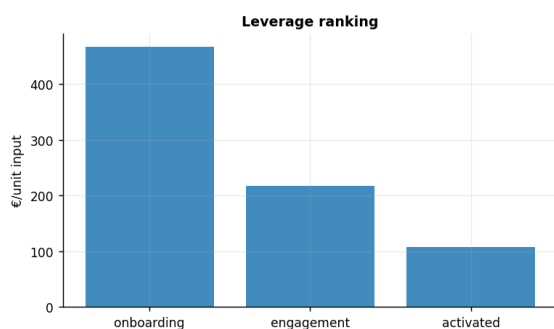


1.5 6 · Decide, in euros — leverage ranking *and* its fragility

Two moves. First, **rank levers** by the € of conversion value from a unit improvement in each stage's input (using the total downstream effect — onboarding's leverage cascades). Second, the honest part: **sensitivity of the mediation split to unmeasured $M-Y$ confounding**. If an omitted trait drives both engagement and conversion, the indirect path is over-credited — we sweep a confounding strength and show how far the “invest in onboarding vs engagement” call would have to be pushed before it flips.

lever	€/unit	P(>0)
onboarding	467.0	1.0
engagement	217.0	1.0
activated	107.0	1.0

Onboarding leads on leverage, and the call is ROBUST: even the strongest $M-Y$ confounding we swept ($\rho=1.5$) doesn't close the €250 gap to engagement.



1.6 7 · Caveats

- **Natural effects lean on strong assumptions** — no unmeasured confounding of any of the three links plus the cross-world condition. Step 6 quantified how much $M\text{--}Y$ confounding would flip the decision; the point estimate alone hides that fragility.
- **Don't control away the mechanism.** The goal is to *decompose* the path; adjusting for a mediator when you wanted the total effect removes the very thing you're pricing (nb 05).
- **Linear-SCM decomposition is exact only if the SCM is linear.** For strong nonlinearities lean on `pathmc`'s simulate-through-the-graph numbers, not hand-multiplied coefficients.